Biax Experiment *(revised. March 2024)*

For Current Calibrations: ~/storage/group/cjm38/default/Calibrations/

Exp. Name: p5986
Operators: ~~ZJ~~ Ke, Yu
Temperature (°C): ~~23~~ 23.2
Relative Humidity: (%): 43.7

Date/Time: 2024-04-15
Hydraulics start: 7556.4
Hydraulics end:
Data Logger Used/Control File: 8ch

Purpose/Description: DDS 10 MPa with Steel "spring" for AM stress drop + 20 PZT for passive acoustics

Sample Block Used and Thickness with **no** Sample:

____ Steel 5x5 cm, _____mm
____ Steel 10x10 cm, _____mm
____ Acoustic Blocks _____

____ SDS Vessel 5x5cm, _____mm
____ AC-DDS Vessel 5x5cm, _____mm
____ Non-AC DDS Vessel 5x5 cm, _____mm

Sample Gouge/Material Preparation:

Material: Barre Gray Granite 15x5
Particle Size, Distribution: # 60 all surfaces
Benchtop Sample Thickness [mm]: _____
Pre-Compaction Sample Thickness [mm]: _____
Post-Compaction Sample Thickness [mm]: _____

	Block 1	Block 2
Empty Block Weight [g]		
Weight of Material Used [g]		
Sample Block Weight [g]		
Weight of Gouge [g]	0.8	0.8

Contact Area [m]: 0.007696

Load cells:

Load cell	*Calibrations (mV/kN)*		*Target stress (MPa)*	*Init. Voltage*	*Volt. @ load*
44 mm H	LC1 LG: 12.894 ~~LC2 LG: 12.545~~	HG: 129.955	*Hor:* *Calibration: 0.09654 (V/MPa)*	-0.449	0.5164 @ 10 MPa
44 mm V	~~LG: 13.074~~	HG: 120.365	*Vert:* *Calibration: (V/MPa)*		

Vessel Pressure:

Pore Fluid: _____

Calibrations (V/MPa)		*Pressures (MPa)*	*Initial Voltage*	*Voltage @ Load*
LG: 0.145403	HG: 1.5238391	*PpA:*		
LG: 0.1461386	HG: 1.5124781	*PpB:*		
Gain : 0.1454517		*Pc:*		
LG: NA	HG: NA	*Pdiff:*		

Displacement Transducers

Horz. DCDT	Long Rod	Short Rod	**Vert. DCDT**	Long Rod
Low Gain	*1.296157 mm/V*	*1.38357 mm/V*	~~Low Gain~~	~~1.845203 mm/V~~
~~High Gain~~	*0.628816 mm/V*	~~0.670257 mm/V~~	High Gain	*0.571599 mm/V*

Horz On-Board DCDT	Biax	Vessel	**Vert. On-Board DCDT**	Biax	Vessel
Low Gain	*0.855923 mm/V*	*0.847759 mm/V*	~~Low Gain~~	~~1.187309 mm/V~~	*0.983013 mm/V*
High Gain	*0.418825 mm/V*	*0.422014 mm/V*	High Gain	*0.590694 mm/V*	*0.495375 mm/V*

Horizontal Servo Settings:
P: 960 D_{atten}: 1
I: 200 Feedback: 1
D: 100 E-gain: 1

Vertical Servo Settings:
P: 960 D_{atten}: 1
I: 200 Feedback: 1
D: 100 E-gain: 1

Chilled water at HPS
1. Temp In (°F): ____
2. Pres. In (psi): ____
3. Temp Out (°F): ____
4. Pres. Out (psi): ____
5. Flow (lpm): ____

Chiller Unit
6.Panel Temp (°F): ____
7. Panel Pres. (psi): ____
8. Near Pres. In (psi): ____
9.Near Pres. Out (psi): ____

Process water at Chiller
10.Temp In (°F): ____
11.Pres. In (psi): ____
12.Temp Out (°F): ____
13.Pres. Out (psi): ____

Hyd. Power Supply (HPS)
14. Tank Temp (°C): _____. 15. Temp. Out (°C): _____. 16. Pres. Out (psi): _____

10,000 <<01>> 1kHz

467,000 H @ 10 MPa

1,005,000 VDCDTs reset/fed

1,044,000 <<02>> 100kHz, verasonics run 1.

1,054,000 Shear @ 10 mm/s

4,957,800 unloaded.

Failed to save strain data